



DiabetesAI

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Background Information

- **Diabetes:** chronic disease that occurs when body has trouble regulating blood sugar levels (Cleveland Clinic)
- In 2021, **38.4 million** people in the US had diabetes (11.6%) with **8.7 million** undiagnosed (3.4%) (CDC)
- An estimated **97.6 million** Americans had pre-diabetes (CDC)
- Over time, high blood glucose levels can have **long-term complications:** risks of heart attacks, strokes, etc. (Cleveland Clinic)
- Early diagnosis can lead to lifestyle changes and more effective treatments, allowing for predictive models for diabetes risks crucial for public health

With our model, you can...



Diabetes prediction

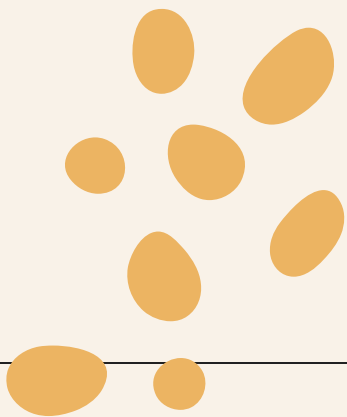


Suggestions for health



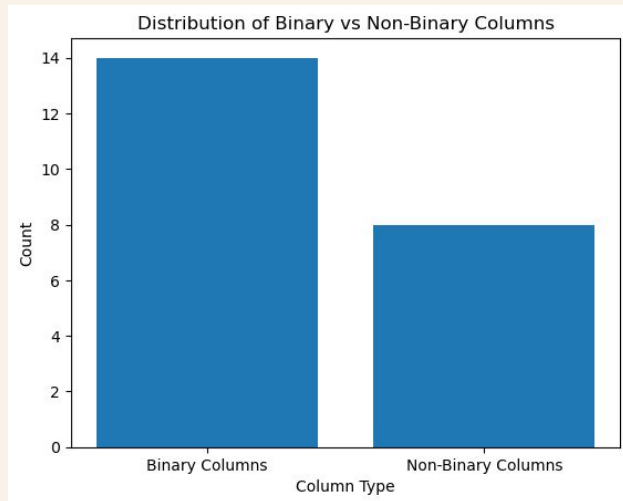
03

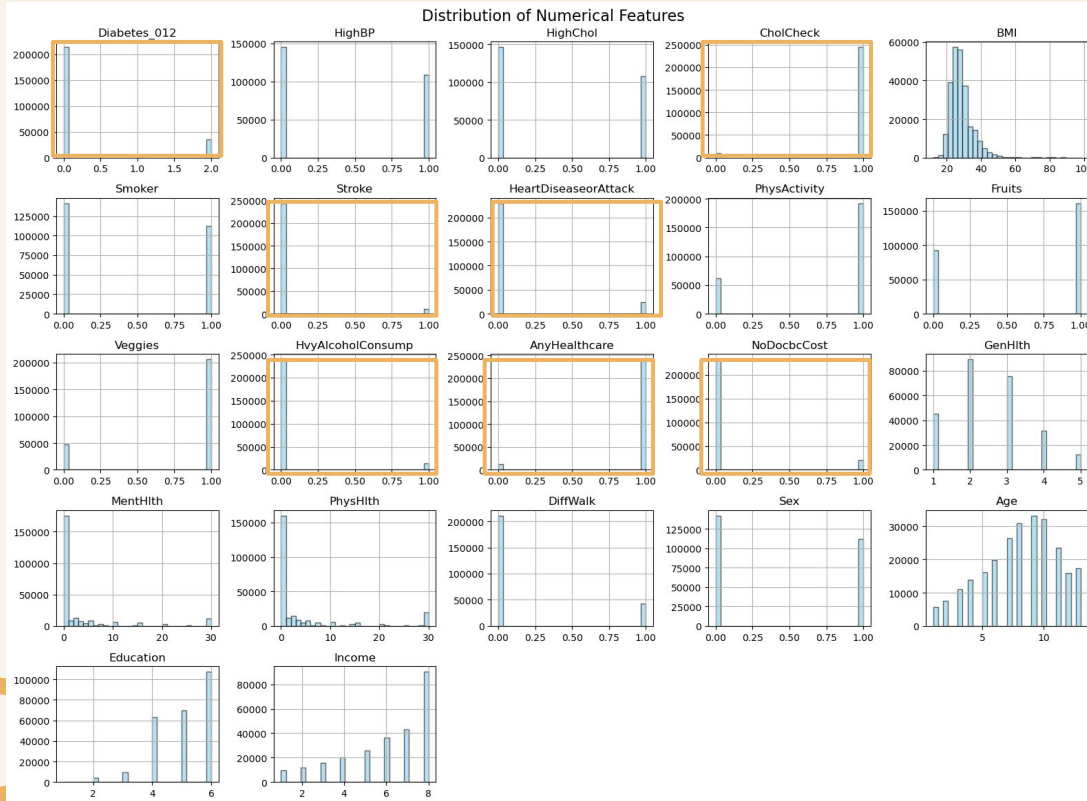
Data Pre-Processing



Dataset Overview

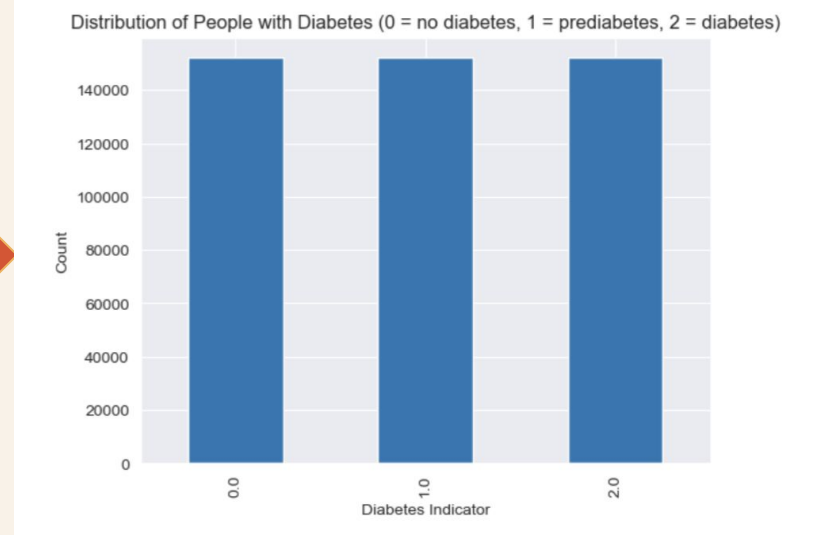
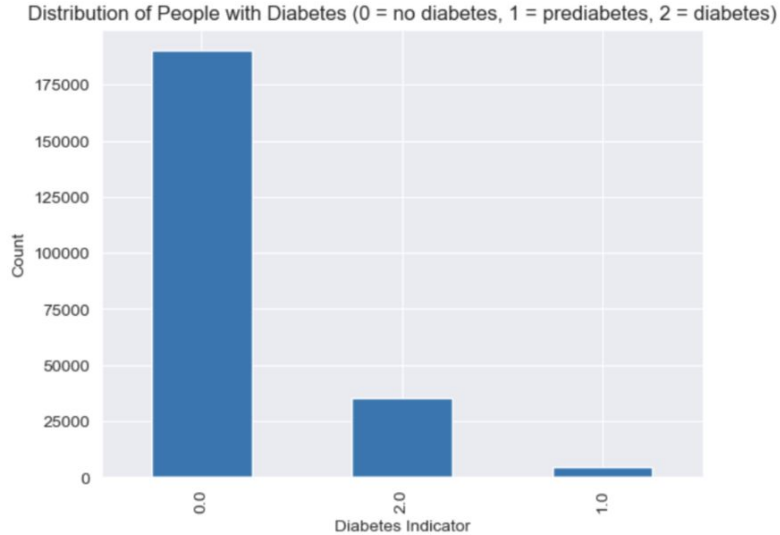
- 253,680 survey responses to CDC's BRFSS2015
- No missing values / categorical values
- Features: **HighChol, PhysAct, Fruits, Smoker, BMI**
- Binary features 14/22
- Target feature (indicator of diabetes)
 - 0 → no diabetes / only during pregnancy
 - 1 → prediabetes
 - 2 → diabetes



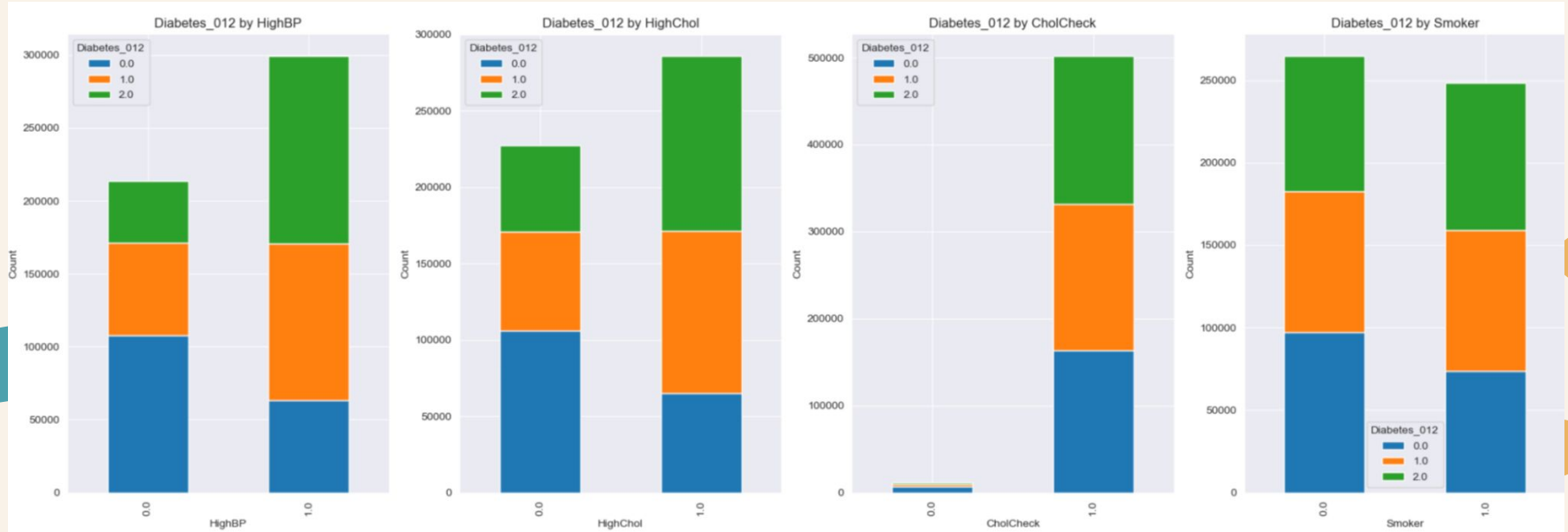


- Imbalanced features
 - Diabetes_012 (target feature)
 - CholCheck
 - Stroke
 - HvyAlcoholConsump
 - AnyHealthcare
 - NoDocbcCost
- Most correlated features
- Most important features

Imbalanced Features

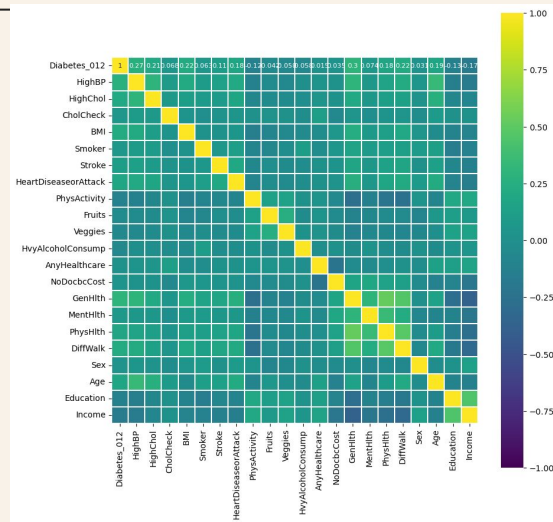


Visualizations



Correlation Analysis

- Diabetes indicator → Diabetes_012
- Most features are correlated with Diabetes_012
- Apply a correlation threshold of 0.05
 - 4 features excluded
 - 5 features negatively related
 - 13 features positively related

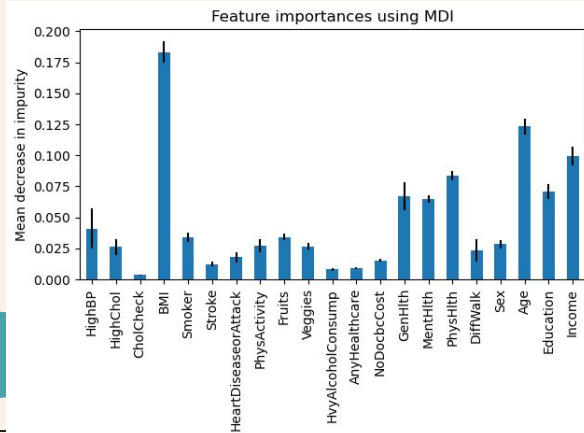


There is 18 strongly correlated values with SalePrice:

```
Diabetes_012      1.000000
GenHlth          0.302587
HighBP           0.271596
BMI              0.224379
DiffWalk         0.224379
HighChol         0.209085
Age              0.185026
HeartDiseaseorAttack 0.180272
PhysHlth         0.176287
Stroke           0.107179
MentHlth         0.073507
CholCheck        0.067546
Smoker           0.062914
HvyAlcoholConsump -0.057882
Veggies          -0.058972
PhysActivity     -0.121947
Education        -0.130517
Income           -0.171483
Name: Diabetes_012, dtype: float64
```

Feature Importance

- Train and test on RandomForest
- Obtain the importance level of each feature by "model.feature_importances_"
- Feature selection based on model performance



	feature	importance
3	BMI	0.183265
18	Age	0.123135
20	Income	0.099255
15	PhysHlth	0.083798
19	Education	0.071117
13	GenHlth	0.067351
14	MentHlth	0.064926
0	HighBP	0.040980
8	Fruits	0.034199
4	Smoker	0.033899
17	Sex	0.028414
7	PhysActivity	0.026898
9	Veggies	0.026693
1	HighChol	0.026252
16	DiffWalk	0.023157
6	HeartDiseaseorAttack	0.017814
12	NoDocbcCost	0.015286
5	Stroke	0.012451
11	AnyHealthcare	0.009149
10	HvyAlcoholConsump	0.008120
2	CholCheck	0.003841



Methods & Results



Model Design



DiabetesAI

Our general model

Diabetes Prediction



Predict if a patient
has diabetes or not



Health Suggestion

Advice for patients
with diabetes

Diabetes Prediction



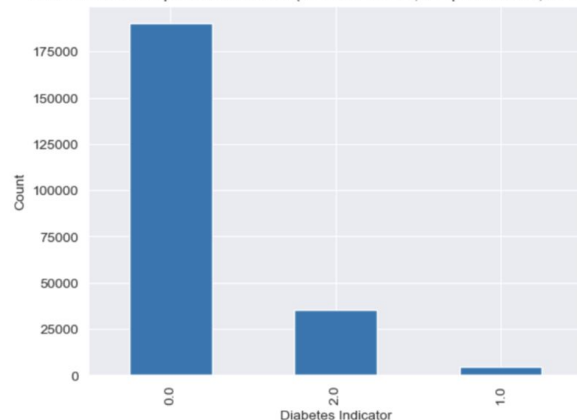
- Use Case:
 - Input: patients' answers for selected features
 - Output: 0/2 → whether this patient has diabetes
- Models to Compare:
 - Logistic Regression
 - Support Vector Machine
 - Random Forest

Results so far

Before resampling:

- Predicts well for non-diabetes group
- No correct prediction for prediabetes
- Low precision score for diabetes

Distribution of People with Diabetes (0 = no diabetes, 1 = prediabetes, 2 = diabetes)

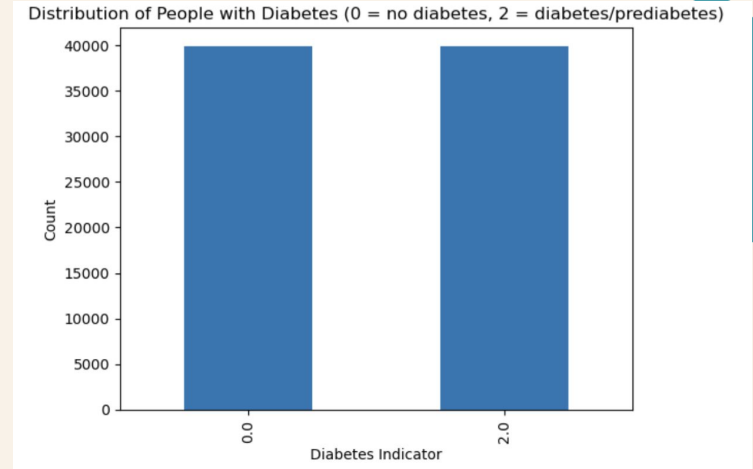


Accuracy: 0.8482734153263954

	precision	recall	f1-score	support
0.0	0.86	0.98	0.92	42795
1.0	0.00	0.00	0.00	944
2.0	0.54	0.18	0.27	6997
accuracy			0.85	50736
macro avg	0.47	0.39	0.40	50736
weighted avg	0.80	0.85	0.81	50736

Data Pre-Processing

- Pre-diabetes combined with diabetes
- Split into test and training set
- Match the size of non-diabetes with diabetes in training set
 - Downsampling
- Scale Features



Results so far

After resampling:

- RandomForest & SVM
- Feature selection
 - Tree based embedded importance
 - RFECV (Recursive Feature Elimination with Cross-Validation)
 - Precision score not improved much

Best model so far:

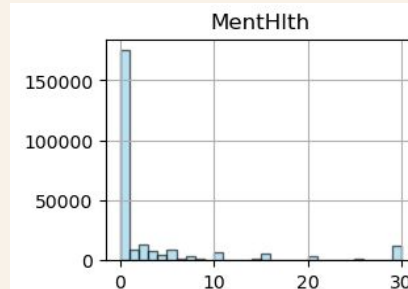
- Logistic Regression
- Feature selection
 - Average coefficient for each feature
 - Filter out Smoker/Fruits/Veggies/PhysActivity/AnyHealthcare/Stroke

Accuracy: 0.7449815521230693

	precision	recall	f1-score	support
0.0	0.75	0.73	0.74	7964
2.0	0.74	0.76	0.75	8027
accuracy			0.74	15991
macro avg	0.75	0.74	0.74	15991
weighted avg	0.75	0.74	0.74	15991

Health Suggestion

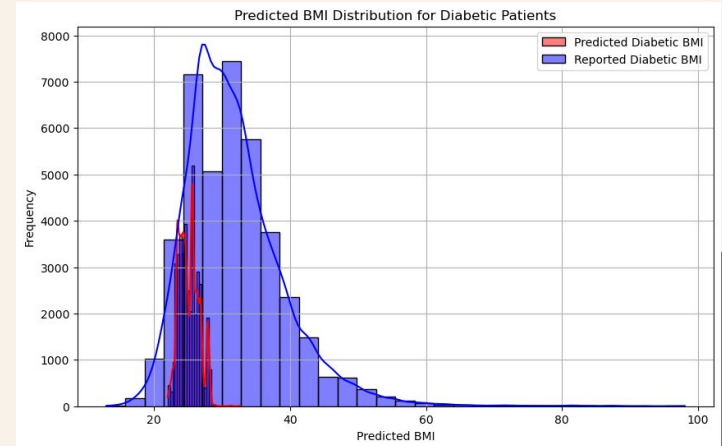
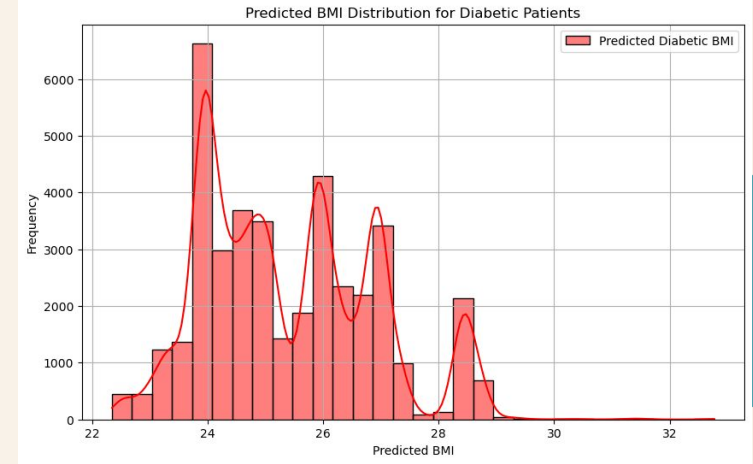
- Dataset
 - patients without diabetes
 - Separate training dataset for predicting highly relevant features to diabetes (BMI/GenHlth)
- Features filtered out:
CholCheck/HvyAlcoholConsump/AnyHealthcare/Stroke
- Input: patients' answers for selected features
- Output: predicted value of (BMI/GenHlth) for the patient
→ suggestion on how to improve



	feature	importance
3	BMI	0.183265
18	Age	0.123135
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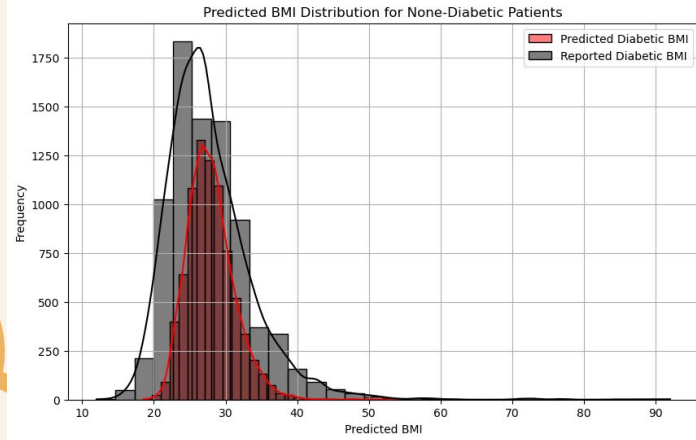
Results so far (BMI)

- BMI
 - body mass index
 - a person's body weight divided by the square of their height
- Random Forest Regressor
 - Train on all healthy subjects (39977)
 - Test set MSE 37.17
- Predict on all the diabetic patients
 - Suggested score centered around 25



Problem & Reflection:

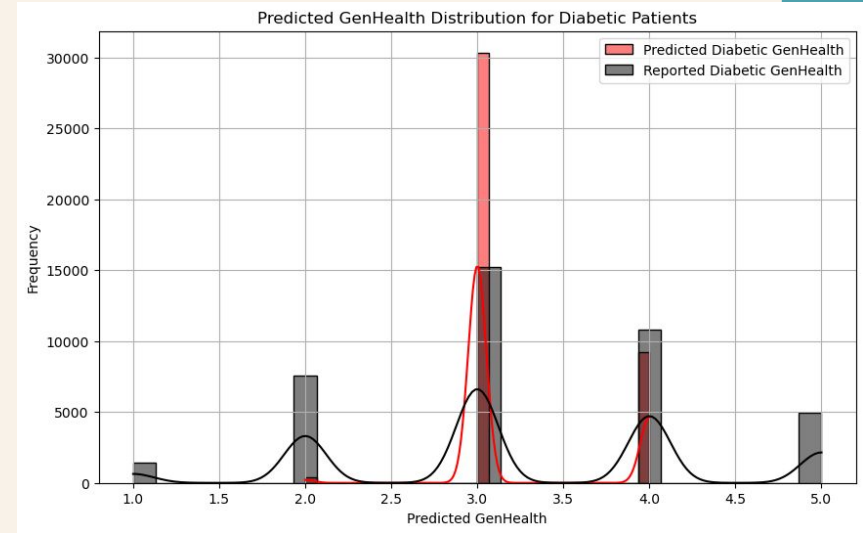
- Model for BMI prediction needs improvement
 - Distribution captured
 - 6 units deviated



	reported BMI	suggested BMI
0	30.0	23.3800
1	25.0	26.8400
2	28.0	23.9800
3	23.0	26.9600
4	27.0	23.7600
5	37.0	27.8900
6	28.0	27.8900
7	27.0	23.3800
8	34.0	24.9325
9	24.0	24.9000

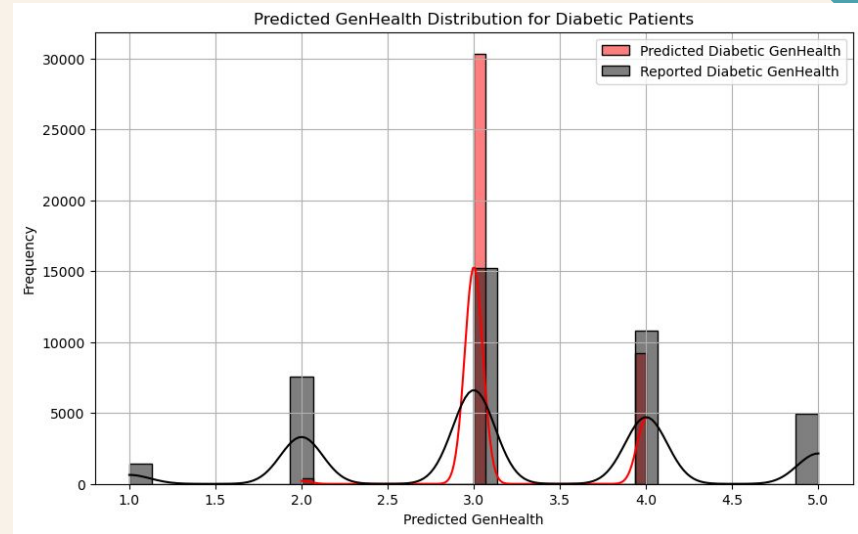
Results so far (GenHealth)

- GenHealth score
 - scale 1-5
 - 1 = excellent; 2 = very good; 3 = good; 4 = fair; 5 = poor
- Random Forest Classifier
 - Train on all healthy subjects
 - Testset MSE 0.89
 - Testset R^2 score 0.16
- Predict on all the diabetic patients
- Suggested score $\rightarrow$ 3



Problem & Reflection:

- General health score too vague
- Need more specific quantification or qualification of health conditions



Model Design



DiabetesAI

Our general model

Diabetes Prediction

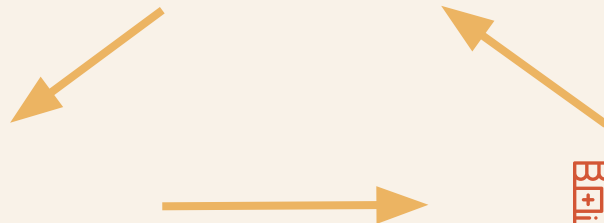


Predict if a patient has diabetes or not

Health Suggestion



Help patient get healthy again



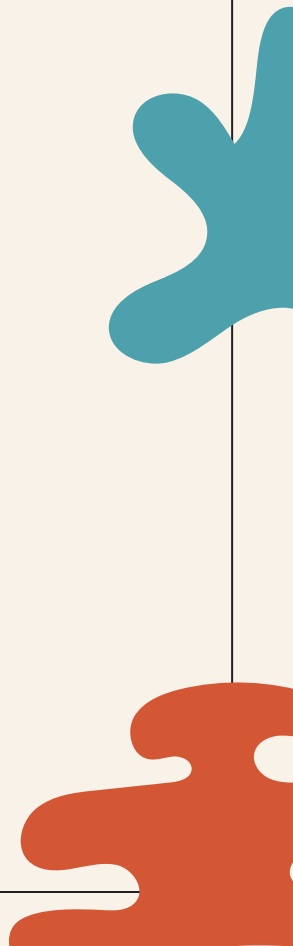
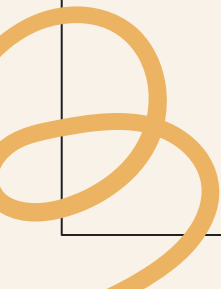
Reflection

- Can survey questions like this provide accurate predictions of whether an individual has diabetes?
 - In general, yes (0.74 precision)
 - Need more data from patients with diabetes
- What factor is most related with the presence of diabetes?
 - BMI
 - GenHealth
 - MenHealth
- During future collection, what kind of data should be focused on specifically?
 - Health conditions → related disease/physiological features
 - Stroke gets filtered out during feature selection
 - Need more data from patients with diabetes



Suggestions for Improvement

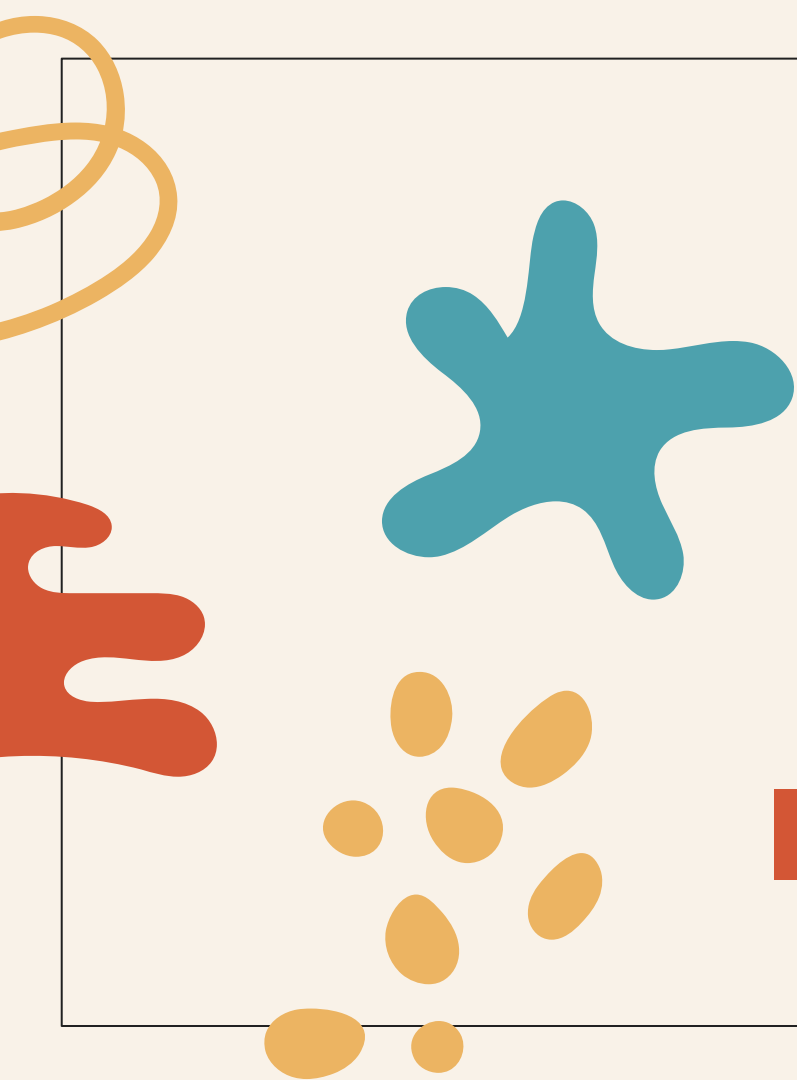
- Addressing Imbalanced Classes
 - SMOTE, class weighting, combination
- Hyperparameter Tuning
- Ensemble Learning
 - Combine multiple models for better generalization on imbalanced data





05

Evaluation Metrics

The slide features several decorative elements: a blue five-pointed star in the upper left, a cluster of yellow oval shapes in the lower left, a red hand-like shape on the far left, and a red horizontal bar below the title. The background is a light beige color.



05

Results



Evaluation Metrics

- Precision, recall & F-1 score
- Confusion matrix (TP/FP)
- ROC-AUC curve

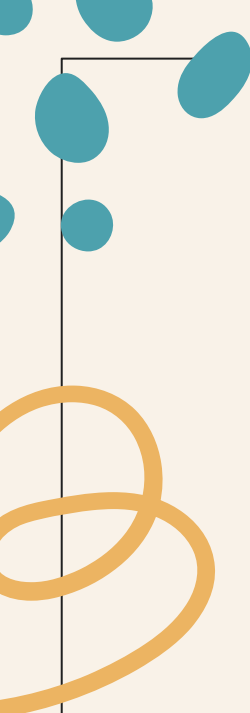


Resources

CDC. "National Diabetes Statistics Report." *Diabetes*, 15 May 2024,
www.cdc.gov/diabetes/php/data-research/index.html.

Dataset:

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>



Q&A